

Ricardo Jaimes Chavez

AI Engineer | Production LLM and Agentic Systems on Structured Financial Data

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SUMMARY

AI engineer building production agentic systems in high-stakes financial domains. Since 2025 in Mexican fiscal compliance, where a wrong output is a legal filing with the tax authority. Shipped solo: a production MCP server (46 tools), a three-tier Claude orchestration layer with cost-aware model routing, and an AI-native compliance platform in private beta with CI-gated evals, end-to-end observability, multi-tenant security, human-in-the-loop approval on irreversible operations, and 8,900+ tests across two repos. Previously Lead Developer at RetailDAO, where I led a smart-contract audit and executed the on-chain governance action that permanently capped the token supply. Python and TypeScript/Node. English-first documentation and async collaboration with North American teams.

EXPERIENCE

AI Engineer and Founder, DISAI (independent) | Jun 2025 - Present

- **Production MCP server (46 tools, 18 prompts, 95 resources).** Designed, built and operate a Model Context Protocol server (TypeScript/Node, Railway) covering the full CFDI 4.0 lifecycle: invoicing, payroll, carta porte, payment complements, PAC stamping and multi-tenant credentials. Migrated to MCP SDK v2 and the stateless 2026-07-28 protocol within a week of its release.
- **Three-tier agentic orchestration on Claude.** Haiku 4.5 router with a deterministic tie-break across 10 fiscal domains, 10 domain agents using native tool use with agentic loops and self-correction, scoped tool allowlists, and cost-aware model routing (Sonnet 4.6, Sonnet 5, Opus 4.8).
- **Reliability under irreversible operations.** Human-in-the-loop confirmation before any tax-authority stamping, exactly-once stamping via in-process single-flight plus an atomic PostgreSQL claim, fail-open vs fail-closed policy per subsystem, and DB-persisted circuit breakers on PAC providers.
- **Multi-tenant security.** PostgreSQL row-level security, short-lived HMAC-signed request assertions per org and tool, encrypted per-tenant certificates, TOTP MFA, Redis-backed rate limiting and a structured auth audit trail.
- **Retrieval and context injection.** Domain-scoped retrieval over MCP resources with live per-domain context injection and fuzzy search across 52k+ SAT catalog entries. Designed the hybrid retrieval upgrade (trigram plus embeddings, RRF) where retrieval only proposes candidates and exact validation commits.
- **Evals, testing and observability.** Routing eval harness gating CI on every pull request; Langfuse tracing with per-session cost attribution and Sentry for errors. 8,900+ tests from unit and contract through integration and Playwright E2E; the MCP server holds 97% line and 90% branch coverage, enforced in CI.
- **Production ownership.** 28 architecture decision records and 8 written postmortems with root cause, blast radius, fix and lesson.
- **Applied mathematics.** Supplier-network risk graph with BFS propagation and DFS cycle detection for tax-fraud carousels; published the decay bound and released it as the open-source efos-risk-graph npm package.
- **Full-stack and delivery.** Next.js 16 / React 19 frontend, Node backend, PostgreSQL and Redis, Zod validation at every boundary. Ran discovery with Mexican accounting firms and turned regulatory requirements into agent workflows, building solo with Claude Code and reviewing all generated code before production.

Core Contributor / Lead Developer, RetailDAO (Base mainnet) | Jun 2024 - Jan 2026

- Led the smart-contract audit that traced an active-mint risk flag to an Aragon plugin retaining execute() permissions; designed and executed the mitigation that made the mint DAO immutable and capped supply at 1B tokens, passed by on-chain token-weighted vote.
- Built a real-time crypto analytics terminal aggregating 8+ sources over WebSockets with Redis caching, rate limiting, exponential backoff and graceful degradation (React, Node.js, Redis, Docker).
- Authored and shipped the full protocol documentation site (Docusaurus, GitHub Pages).

Data and Research Analyst, Coinsider | Mar 2024 - Sep 2025

- Delivered 25+ peer-reviewed analytics reports (1,200+ pages) on investor risk, market structure and macro forecasting, reaching 70k+ views. Hired after a multi-stage technical assessment.

Data Analysis Consultant, Chainlabs (short engagement) | Jun 2025

- Delivered Python wallet-analysis notebooks for a European crypto-intelligence firm: OSINT, cross-chain data wrangling and relational graphs (Plotly, Matplotlib).

Earlier background

Self-directed work in crypto and decentralized systems from 2020 (protocol mechanics, on-chain operations, quantitative market analysis), after 12+ years as a graphic designer, illustrator and tattoo studio owner.

SKILLS

AI / LLM: LLM application development, AI agents, multi-agent orchestration, agentic loops, prompt engineering, evals, hallucination control, human-in-the-loop, cost-aware model routing, retrieval and context injection, Claude API, tool use / function calling, Model Context Protocol (MCP)

Languages and backend: Python, TypeScript, Node.js, SQL, REST APIs, Server-Sent Events, Streamable HTTP, object-oriented design

Data and infrastructure: PostgreSQL, SQLite, Redis, Docker, WebSockets, caching, rate limiting, circuit breakers, idempotency

Security: multi-tenant isolation (row-level security), HMAC request signing, TOTP MFA, encrypted credential storage, audit logging

Frontend: React, Next.js, Tailwind CSS, shadcn/ui

DevOps and practices: CI/CD (GitHub Actions), Git, Railway, Vercel, Vitest, Playwright, Zod, Langfuse, Sentry, ADRs, postmortems, technical documentation

AI-assisted engineering: Claude Code, Gemini CLI, OpenCode (daily production use), AI-driven code review

SELECTED PROJECTS

- **SAT-MCP Server:** production MCP server for Mexican fiscal compliance; 46 tools, 95 resources, 9 in-chat MCP Apps; 6,400+ tests.
- **DISAI-Conta:** AI-native fiscal compliance platform for accounting firms, in private beta; router plus 10 domain agents, CI-gated routing evals, 2,500+ tests.
- **AI Reliability and Evaluation Lab:** open-source, reproducible benchmark comparing direct Sonnet 5 with routed Haiku/Sonnet agents, with hashed datasets and CI spend ceilings.
- **Open-source libraries:** efos-risk-graph, agentic-tool-loop, llm-cost-router, mcp-tool-idempotency.
- **Technical writing:** engineering write-ups on model routing, MCP throughput and risk-graph math at tripl3.dev/writing.

EDUCATION AND CERTIFICATIONS

- B.A. Graphic Design, Universidad Autonoma de Aguascalientes (2013)
- MIT OCW 6.1200J, Mathematics for Computer Science (self-study, in progress)
- Anthropic (2025): Model Context Protocol: Advanced Topics; Introduction to Model Context Protocol; Claude Code in Action; AI Fluency: Framework and Foundations
- DataCamp: Building Scalable Agentic Systems (2025); n8n: Self-Hosted for Enterprises (2025)

LANGUAGES

Spanish: native | English: professional working proficiency